

Dr Aras Selvi

Assistant Professor of Operations & Technology



Bio

Aras is an Assistant Professor of Operations Research at UCL School of Management. He is interested in decision making under uncertainty and its applications in machine learning, privacy, and fairness. In his recent works, he has been working on designing **optimal privacy mechanisms** for AI applications, developing efficient algorithms for **robust machine learning**, as well as developing data-driven approaches to **fair resource allocation** in high-stakes domains.

Aras' research is published at leading operations management journals, including *Operations Research* and *Manufacturing & Service Operations Management*, applied mathematics journals, including *Mathematical Programming* and *INFORMS Journal on Computing*, as well as major AI conferences, including *Neural Information Processing Systems* and *International Conference in Machine Learning*.

Before joining UCL, Aras was a postdoctoral research associate at the Department of Operations Research and Financial Engineering at Princeton University. He received his PhD from Imperial College London as a member of the group *Models and Algorithms for Decision-Making under Uncertainty*. He has also held affiliations with JP Morgan AI Research, The Alan Turing Institute, Imperial Computational Optimization Group, Imperial Business Analytics, and Data Science Institute.

His doctoral thesis, “**Models and algorithms for safeguarded data-driven decision making**”, was selected as the best quantitative thesis at Imperial College London by the quantitative research firm G-Research, best thesis in the UK by the Operational Research Society, as well as one of the four international finalists at the George B. Dantzig Dissertation Award. His papers received several awards and recognitions from the Institute for Operations Research and the Management Sciences.

In addition to his theoretical research agenda, Aras enjoys working with industry and government partners for applied research collaborations as well as delivering executive education. His current collaborations include a partnership with the technology intelligence platform *Dragonfly* to “develop next-generation AI systems that help businesses make safer, more effective technology decisions”, and with the *United Nations World Food Programme* and *Columbia Global Centers* to “develop reliable AI tools for disaster relief that manage uncertainty end-to-end, transforming noisy, chaotic request streams into actionable recommendations for governments”.



Expertise

- Decision making under uncertainty
- Reliable AI Optimization
- Data-driven decision making
- Building machine learning algorithms
- Data analytics Coffee ☕

Academic Background



PRINCETON
UNIVERSITY

IMPERIAL

The
Alan Turing
Institute

Relevant Teaching Experience

- Data Visualisation and Storytelling { London Business School }
- Big Data, AI & Machine Learning { Imperial Business School }
- Algorithmic Techniques in ML { London School of Economics }

🏆 LSE Teaching Award

Recent Grants

- \$1M Innovate UK grant with Dragonfly for **next-gen safe AI** systems.
- \$150K Columbia Global grant to develop **resilient governmental AI**.

Industry Research & Collaboration

J.P.Morgan



KPMG



Public & Industry Talks

- 📢 Pint of Science
- 📢 AI UK
- 📢 Uncertainty in AI
- 📢 McKinsey QuantumBlack
- 📢 BCG X
- 📢 Legal & General